Description of Malignancy Rates in Childhood- and Adult-Onset Systemic Lupus Erythematous by Proportional Meta-analysis

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Objective: The aim of this study was to describe the malignancy rates in childhood and adult-onset systemic lupus erythematous (SLE) by proportional meta-analysis.

Methods: Two reviewers screened data from PubMed (1966–2015), EMBASE (1980–2015), and LILACS (1982–2015) for SLE-associated malignancy. Proportional meta-analysis with a random-effects model and 95% confidence intervals (CIs) were calculated according to SLE onset age and mean follow-up time. Statistical difference was defined by 95% CI overlap.

Results: Malignancy rate reported in 30 case series with 96,578 subjects was 3.4% (95% CI, 0.0260–0.0442; $I^2 = 97.6\%$; $P < 0.0001$). It was 4.2% (95% CI, 0.0318–0.0531; $I^2 = 98\%$; $P < 0.0001$) in 25 adult-onset SLE series, compared with 0.5% (95% CI, 0.0003–0.0154; $I^2 = 62.6\%$; $P = 0.03$) in 5 childhood-onset SLE series. There was a significant difference in the malignancy rates, with lower rates in childhood-onset compared with adult-onset SLE. Overall, in those with less than 5 years' follow-up, it was 2.8% (95% CI, 0.013–0.047; $I^2 = 91\%$; $P < 0.0001$) compared with 3.6% (95% CI, 0.0226–0.0531; $I^2 = 98.3\%$; $P < 0.0001$) in those with more than 5 years' follow-up, which was not significant, with 95% CI overlap.

Conclusions: The meta-analysis indicated lower malignancy rates in pediatric-onset SLE compared with adult-onset SLE, but accrued data from childhood-onset SLE are still needed.

Key Words: cohort studies, malignancy, meta-analysis, systemic lupus erythematosus

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There is increasing concern of malignancy susceptibility in systemic lupus erythematosus (SLE). Risk estimates have more often been calculated in case series and population cohorts,[1,2] including subjects with definite diagnosis of SLE according to the American College of Rheumatology 1997 criteria[3] or previous American Rheumatism Association 1982 criteria.[4] More recently, collaborative work reported by SLICC (Systemic Lupus International Collaborating Clinics) group was published.[5] Estimated standardized incidence ratio, which is the ratio of observed and expected malignancy rates in SLE patients compared with reference population, was also reported.

Other question of interest is the comparative incidence of hematologic and nonhematologic malignancy rates.[6–9] Associated lifestyle risks factors, such as smoking, obesity, viral exposure, and immunosuppressive treatment; or genetic factors; or race or ethnicity could account for increased relative risks. Recent data confirmed that lymphoma incidence is higher in SLE patients, but the risk factors of such association are still unknown.[10]

Studies on adult-onset SLE confirmed increased risk of non-Hodgkin lymphoma, lung, vulvar/vaginal, and thyroid and decreased risk of hormone-sensitive breast and prostate cancer. The higher risk of vulvar/vaginal site is related to abnormal virus clearance such as human papillomavirus. The increased risk of non Hodgkin lymphomas is related to multiple mechanisms, including viral reactivation and cytokine up-regulation.[8,11,12] Bernatsky et al.[13] addressed the issue of SLE disease activity versus treatment, suggesting cyclophosphamide exposure and high cumulative steroid doses were associated to higher risk, whereas disease activity itself was not clearly associated with malignancy. Overall, reports of malignancy in SLE subjects vary from 1.1% to 2.6%.[5] There were reports from multisite international cohorts from more than 30 centers.

There is also increased malignancy risk associated with childhood-onset SLE. The incidence rates are higher compared with the reference population of those not affected by SLE.[14,15] Still, it is not known if time of disease exposure or similar concurrent risk factors could be implicated. Major concern for both pediatric and adult individuals could be related to cytotoxic or biologic treatment. However, there is no previous study comparing cancer incidence in childhood and adult-onset SLE populations.

Therefore, we revised the reports of malignancy associated with SLE, comparing childhood and adult-onset SLE, by proportional meta-analysis of cohort studies.

METHODS

This report complies with the MOOSE (Meta-analysis of Observational Studies in Epidemiology) statements and quality-control guidelines of prognosis studies.[16,17] Studies were included if participants had an established diagnosis of SLE by clinical evaluation and classification criteria,[3,4] regardless of age, sex, or exposure to steroids or immunosuppressive or biologic treatment.

This review included all longitudinal studies with malignancy reports or a proper score for the SLICC/American College of Rheumatology Damage Index (SDI). The SDI scores 1 point for any biopsy-proven malignancy from each site, excluding dysplasia, after 6 months of SLE onset.[18] Those longitudinal studies were mainly cohort studies with a minimum 6 months' follow-up and cancer registry reports. We excluded reviews and cross-sectional and case-control studies. The primary outcome was any type of malignancy, regardless of topography.

Search strategy for study identification included searches on PubMed (Ovid) (1966 to November 2015), EMBASE (Ovid) (1980 to November 2015), and Latin American and Caribbean Health Sciences Literature (LILACS), via BIREME (1982 to

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November 2015). We selected the articles by a search strategy choosing the terms "lupus" and "cancer" (Appendix 1). There was no language restriction, and reference lists of the identified relevant studies were scrutinized for additional citations.

Two of the authors reviewed the articles independently, to extract the data. A standard form was used for data collection: authors, year of publication, number of participants, subjects mean age, follow-up duration, and description of the outcomes of interest. Studies with incomplete data about the outcome were excluded from the review. Ethical approval was stated in each of the selected studies. We evaluated the risk of bias of the selected articles using a modified version of the Ottawa-Newcastle instrument.[19] It includes confidence in assessment of exposure and outcome and adjusted analysis for differences between groups in prognostic characteristics or missing data.

Statistical Analysis

Malignancy rates were treated as dichotomous variables with 95% confidence interval (CI). The meta-analysis has 3 main steps: test whether the studies results are homogeneous, obtain a global index about effect magnitude of homogeneity or heterogeneity, and are joined to a CI and its statistical significance. For assessing heterogeneity, there can be 2 sources of variability. First, there is the variability due to sampling error, also named within-study variability. The sampling error variability is always present in a meta-analysis because every single study uses different samples. The other source of heterogeneity is the between-studies variability, which can appear in a meta-analysis when there is true heterogeneity among the population effect sizes estimated by individual studies. We calculated statistical heterogeneity by I^2 statistics. Significance was assumed when I^2 was greater than 50%. This reflects the percentage of the variability in effect estimates resulting from heterogeneity rather than sampling error.[20] Assessing heterogeneity in meta-analysis is essential because the presence versus the absence of true heterogeneity (between-studies variability) can affect the statistical model the meta-analyst decides to apply. When studies' results differ by the sampling error (homogeneous case), a fixed-effects model can be applied to obtain an average-effect size. By contrast, if the study results differ by more than sampling error (heterogeneous case), then the meta-analyst can assume a random-effects model, in order to take into account both within- and between-studies variability.

Because of the clear differences of results among the included studies and several uncontrolled variables, we used a random-effects model[21] to perform the proportional meta-analysis. The software used was StatsDirect (StatsDirect Ltd, Altrincham, Cheshire, United Kingdom).

Statistical difference between studied groups was considered significant if their combined 95% CIs did not overlap. We considered $P < 0.1$ as significant for the calculation of I^2 heterogeneity statistic.[22,23]

We performed a subgroup analysis according to different SLE onset ages, defining childhood-onset subjects at 18 years or younger compared with those older than 18 years at SLE onset. Mean follow-up time (≤5 vs. >5 years) subgroups were also compared.

Data summary is presented in forest plots. Each horizontal line of the forest plot represents 1 cohort study included in the meta-analysis. The length of the line corresponds to 95% CI effect estimation in that case series. The effect estimates are marked by a solid black square. The size of the square represents the weight that the corresponding study exerts in the meta-analysis. Pooled estimated rates are marked with an unfilled diamond at the bottom of the forest plot. Confidence intervals are displayed as horizontal lines through the diamond; this line might be contained within the diamond if the CI is narrow.

RESULTS

This literature search identified approximately 2726 titles (Fig. 1). After exclusion of duplicate records and studies that did not meet our inclusion criteria, 56 studies were examined in full text for qualitative analysis. Results of 30 articles with a total of 96,578 subjects were included in the meta- analysis.

Population characteristics by country, number of participants, and follow-up time are presented in the Table 1. Among the included studies, 23 were from Western countries (United States, Canada, Latin America, and Europe),[5,9,15,24–42] whereas 5 studies were from eastern countries (Hong Kong and Korea),[43–47] and 2 studies were from Middle East (Saudi Arabia and Israel).[48,49] Sample size in each of the studies ranged from 47 to 30,478 subjects, with females being the majority.[27,46] Only 2 studies reported specific types of malignancy, breast and lung cancer, respectively[9,32]; the remaining reported overall malignancy rates (Table 1).

Most of included studies matched exposed and nonexposed participants for the variables associated with the outcome, or the authors adjusted the statistical analysis for prognostic variables. Overall, included studies were rated with low risk of bias for the following domains: selection of exposed and nonexposed cohorts

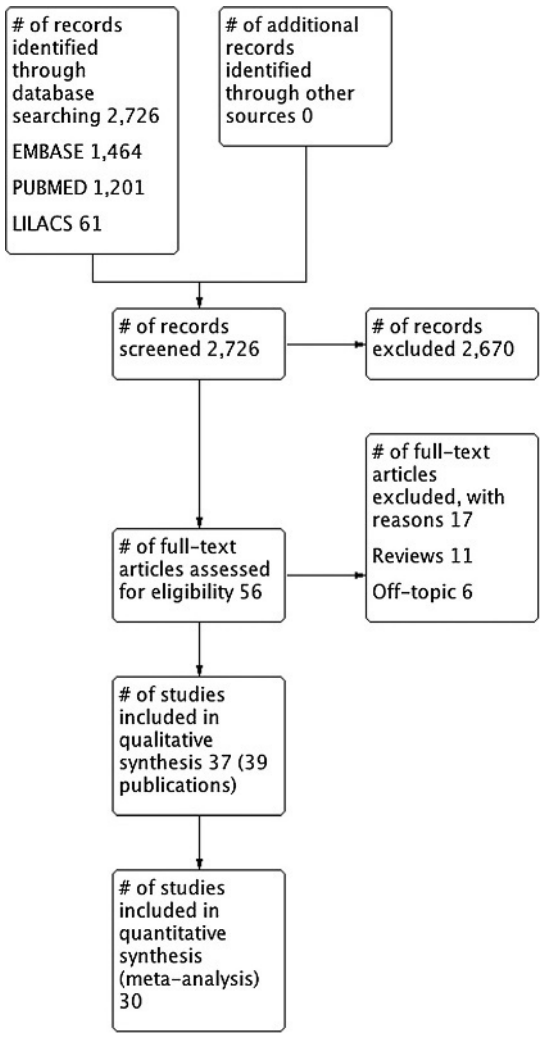

FIGURE 1. MOOSE flowchart.

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TABLE 1. Population Characteristics of Included Cohorts

Reference	SLE, n	Malignancy, n	Sex (Female/Male), n	Period of Recruitment	Country	SLE Onset Age (>18 and <18 y)	Mean SLE Follow-up, y
Khaliq et al.[9] (2015)	18,423	416	416/0	2006	United States	>18 y	—
Chang et al.[43] (2014)	1052	53	—	2000–2012	Korea	>18 y	—
Dey et al.[24] (2013)	595	36	—	1978–2010	United Kingdom	>18 y	15
Hidalgo-Conde et al.[25] (2013)	175	5 (9)	—	1990–2006	Spain	>18 y	5
Bernatsky et al.[5] (2013)	16,409	644	—	1958–2009	International	>18 y	7
Bernatsky et al.[13] (2013)	1020	14	837/183		United States, Canada	≤18 y	8
Dreyer et al.[26] (2011)	576	32	508/68	1990–2006	Denmark	>18 y	13
Tam et al.[44] (2011)	137	12	137/0	—	Hong Kong	>18 y	3
Chen et al.[45] (2010)	11,763	259	—	1996–2007	Taiwan	>18 y	5
Parikh-Patel et al.[27] (2008)	30,478	1273	—	1991–2002	United States	>18 y	5
Al-Mayouf and Al Sonbul[48] (2008)	89	0	76/13	1995–1997	Saudi Arabia	≤18 y	5
Nossent et al.[28] (2007)	915	67	—	2000–2004	Europe	>18 y	5
Lee et al.[46] (2007)	47	0		1988–2004	Hong Kong	≤18 y	7
King and Costenbader[29] (2007)	2020	325	—	1992–2005	United States	>18 y	18
Cassano et al.[30] (2007)	197	1	—	—	Argentina	>18 y	10
Guarize et al.[31] (2007)	60	1	58/2	2001	Brazil	>18 y	6
Bin et al.[32] (2007)	9547	62	—	—	International	>18 y	8
Ruiz-Irastorza et al.[33] (2007)	235	13	209/26	—	Spain	>18 y	10
Bandeira et al.[34] (2006)	57	0	44/13	—	Italy, Brazil	≤18 y	4
Becker-Merok and Nossent[35] (2006)	158	3	—	—	Norway	>18 y	12
Bezerra et al.[36] (2004)	82	0	77/5	1998–1999	Brazil	>18 y	3
Cervera et al.[38] (2003)	1000	23	—	1990	International	≤18 y and >18a	10
Gladman et al.[39] (2003)	73	1	—	1970–2003	Canada	>18 y	15
Ravelli et al.[50] (2003)	387	0	—	1999	International	≤18 y	5
Sundaramurthy et al.[37] (2003)	87	3	—	1995–2001	United States	>18 y	5
Molad et al.[49] (2002)	151	7	—	1989–1998	Israel	>18 y	4
Maddison et al.[40] (2002)	141	9	—	—	International	>18 y	5
Cibere et al.[41] (2001)	247	27	—	1975–1994	Canada	>18 y	12
Mok et al.[47] (1999)	252	2	201/51	—	Hong Kong	>18 y	9
Pettersson et al.[42] (1992)	205	17	182/23	1967–1987	Finland	>18 y	20

(−) not reported.

from the same population, confidence in the assessment of exposure (SLE), outcome (malignancy), and follow-up time.

Malignancy rate in 30 case series with 96,578 subjects with SLE was 3.4% (95% CI, 0.0260–0.0442). There was a significant heterogeneity ($I^2 = 97.6\%$) showing the consistency of clinical and methodological aspects between the studies included in the meta-analysis ($P < 0.0001$) (Fig. 2).

Malignancy rates in 25 adult and 5 childhood-onset SLE series were 4.2% (95% CI, 0.0318–0.0531) and 0.5% (95% CI, 0.0003–0.0154), respectively. There was a significant difference comparing the groups according to SLE onset age, as the 95% CIs did not overlap (Fig. 2). Also, statistical heterogeneity was significant: adult-onset SLE $I^2 = 98\%$, $P < 0.0001$, and childhood-onset SLE, $I^2 = 62.6\%$, $P = 0.03$.

Reported follow-up was grouped as mean follow-up time of less than or equal to 5 years and more than 5 years. Malignancy rate from 11 studies with a total of 34,462 SLE subjects with mean follow-up of less than or equal to 5 years was 2.8% (95% CI, 0.013–0.047), whereas the rate in 16 studies with a total of 30,724 SLE subjects, with mean follow-up time more than 5 years, was 3.6% (95% CI, 0.0226–0.0531). Follow-up time was not reported in 2 of the reported studies, comprising 19,475 SLE subjects (Table 1). There was no significant difference between these groups as the CIs overlap. Also, I^2 heterogeneity was significant in those with less than 5 years' and more than 5 years' mean follow-up time, $I^2 = 91\%$, $P < 0.0001$, and $I^2 = 98.3\%$, $P < 0.0001$, respectively.

This meta-analysis pointed to a significantly higher malignancy rates in adult-onset compared with childhood-onset SLE.

DISCUSSION

The main findings of malignancy reports in adult- and childhood-onset SLE cohorts, from different geographic regions and population sources, were combined in the meta-analysis. We reviewed more robust data in adult SLE population compared with only 5 childhood-onset SLE series reporting malignancy, to date. This included a large multisite registry from 10 pediatric

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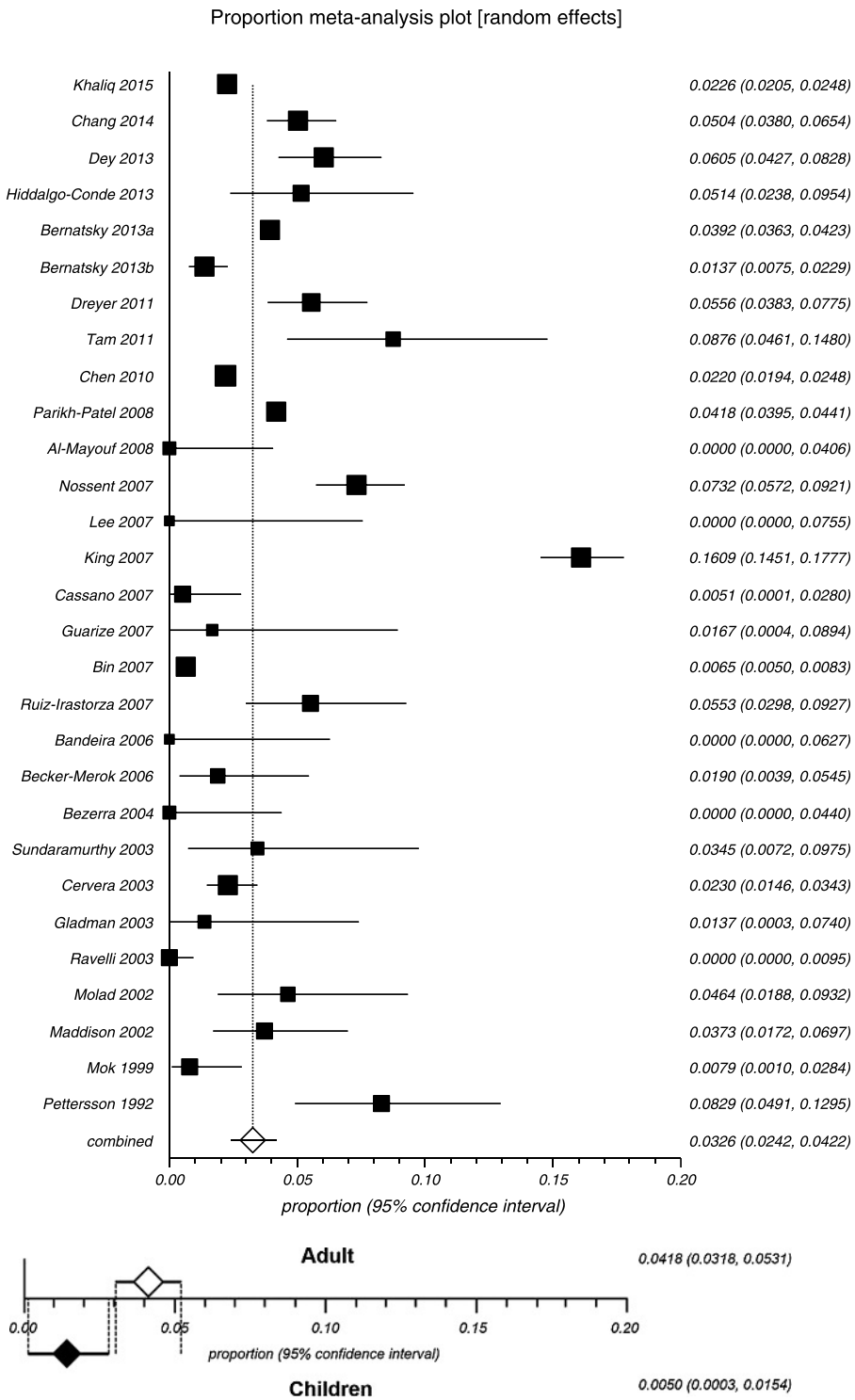

FIGURE 2. Malignancy rates reported in childhood-onset versus adult-onset SLE cohorts (I^2 = 97.7%).

rheumatology centers in United States and Canada[15] and 4 smaller series discriminating malignancy diagnoses during long-term follow-up.[34,46,50]

Systemic lupus erythematous onset before the age of 18 years is remarked by more severe presentation compared with adult-onset SLE. Considering all the differences in disease activity, organ involvement, therapeutic choices, and malignancy risk as comorbidity, there was lower incidence of malignancy in childhood-onset SLE compared with adult cohorts; however, only a few reports were published to date.

Our study limitations were the wide variation of studies compiled and limited reports about age range stratification in

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childhood-onset and adult-onset SLE, malignancy site, and organ involvement comparisons, as well as comorbidity additional risk factors. There was also a wide methodological variation with only 1 inception cohort[24] and a few healthy population–controlled studies.[9,25,43] Most of included population studies were reports of the National Cancer Registry matched to hospital discharge diagnoses as case-finding strategy, but smaller SLE series, from rheumatology practices, reporting damage index (SDI) were also pooled in the meta-analysis. The proportion of cases could be also a source of bias, but the search strategy and study selection allowed inclusion of all the published pediatric series with long-term follow-up, in order to power the analysis.

This systematic review adds to prior work about malignancy risk in SLE patients by comparing different groups according to SLE onset age, including childhood-onset and adult-onset subjects with the same disease, although with a much lower proportion of pediatric subjects, because of lower frequency of childhood-onset SLE. This review may be limited by asymmetry in the comparison; therefore, we could not rule out bias possibility because of unbalanced pooled proportions.

There are prior studies, including recent meta-analyses,[11,51–53] addressing specific malignancy site or organ involvement in adults, such as breast, ovarian, endometrial,[53–55] or urologic cancer.[6,7] In addition, there are others reporting leukemia and lymphoma,[8,12,13] but we did not perform the site-specific malignancy analysis because of scarce data in childhood-onset SLE subjects. In this group, only overall malignancy rates were reported. Previous reviews and meta-analysis pointed out to a higher risk of non-Hodgkin lymphoma, estimated from overall malignancy rates, but so far no pooled rates were estimated in childhood-onset SLE.

In the only population study of childhood-onset SLE comparing the malignancy standardized incidence ratio, it was 4.7% (95% CI, 2.6%–7.8%), higher than that for the reference population. In that study, most events occurred 10 to 19 years after SLE diagnosis. This study reports age group stratification, sex, and disease duration, varying from less than 1 year to more than 20 years. Malignancy presentation was represented by acute leukemia and lymphoma. The younger patients presented pancytopenia related to leukemia, whereas older patients presented either Hodgkin or non-Hodgkin lymphoma.[15] Very little is known about malignancy risks in other pediatric rheumatic diseases, such as juvenile idiopathic arthritis, dermatomyositis, or Sjögren syndrome.[14] In adult-onset SLE, the higher malignancy risk in comparison to the general population occurs in those younger than 45 years.[5]

Despite all the limitations, malignancy surveillance in childhood-onset SLE is needed. The background risk must be evaluated for safe treatment with glucocorticoids and immunosuppressive and biologic treatment. It is indeed a concern from patients, parents, physicians, and other health professionals caring for children with pediatric rheumatic diseases. Long-term prognosis studies must be planned to assess outcome in SLE beyond survival.

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APPENDIX 1. Search strategy

(systemic lupus erythematosus OR Lupus Erythematosus Disseminatus OR Libman-Sacks Disease OR Libman Sacks Disease) **AND** (Neoplasms OR Neoplasm OR Tumors OR Tumor OR Cancer OR Cancers OR Benign Neoplasms OR Benign Neoplasm OR malignancy OR malignancies OR Hematological Neoplasms OR Hematological Neoplasm OR Hematologic Neoplasm OR Hematological Malignancies OR Hematological Malignancy OR Hematologic Malignancy OR Hematologic Malignancies OR Hematopoietic Neoplasms OR Hematopoietic Malignancies OR Hematopoietic Malignancy OR Hematopoietic Neoplasm OR blood cancers OR liquid tumors OR solid tumors OR solid tumor OR solid neoplasm OR solid neoplasms OR solid cancers OR solid cancer OR organ tumors OR organ tumor) **AND** (incidence[MeSH:noexp] OR mortality [MeSH Terms] OR follow up studies[MeSH:noexp] OR prognos*[Text Word] OR prognosis OR prognostic OR prognostics OR predicts OR risk factor OR risk factors OR predict*[Text Word] OR course*[Text Word]) OR prognosis*[Title/Abstract] OR (first[Title/Abstract] AND episode [Title/Abstract]) OR cohort[Title/Abstract]) OR (controlled trial$*) OR (randomised controlled trial$*)

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